



PIANO KEY

DETECTION





1

What Problem

We wanted to try to detect piano keys being played through computer vision

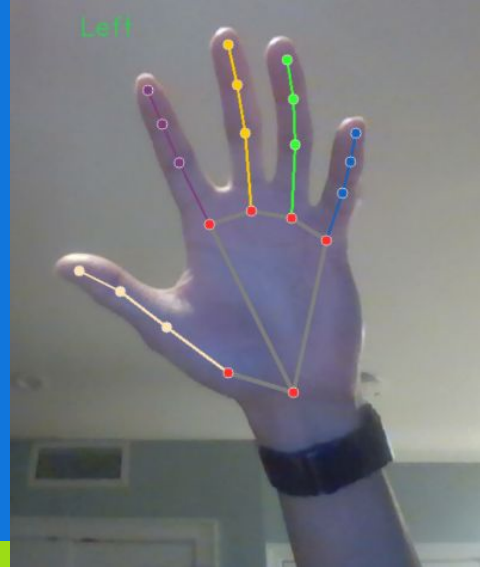
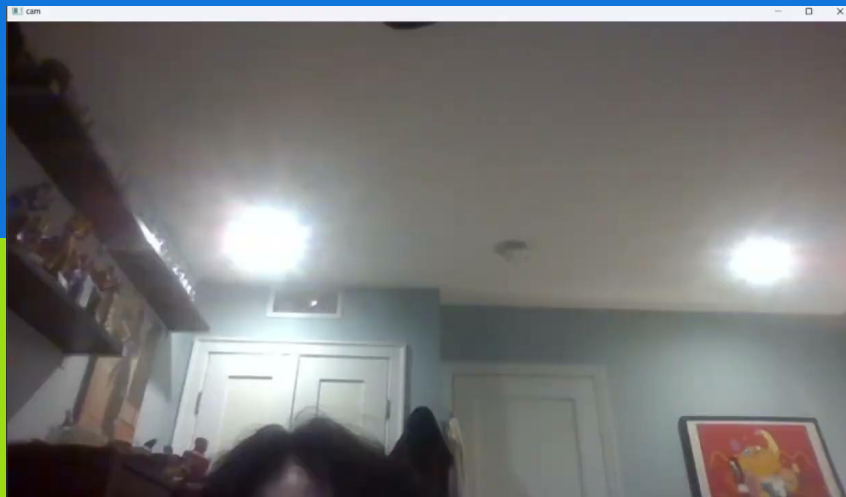
2

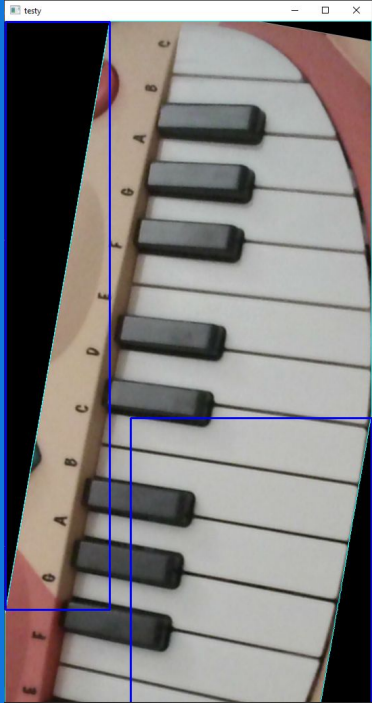
Why solve this problem

There aren't many computer science softwares that utilized musical instruments.

A practical problem for beginner musicians is not having access to sheet music to a specific song but seeing another person play it

MEDIAPIPE





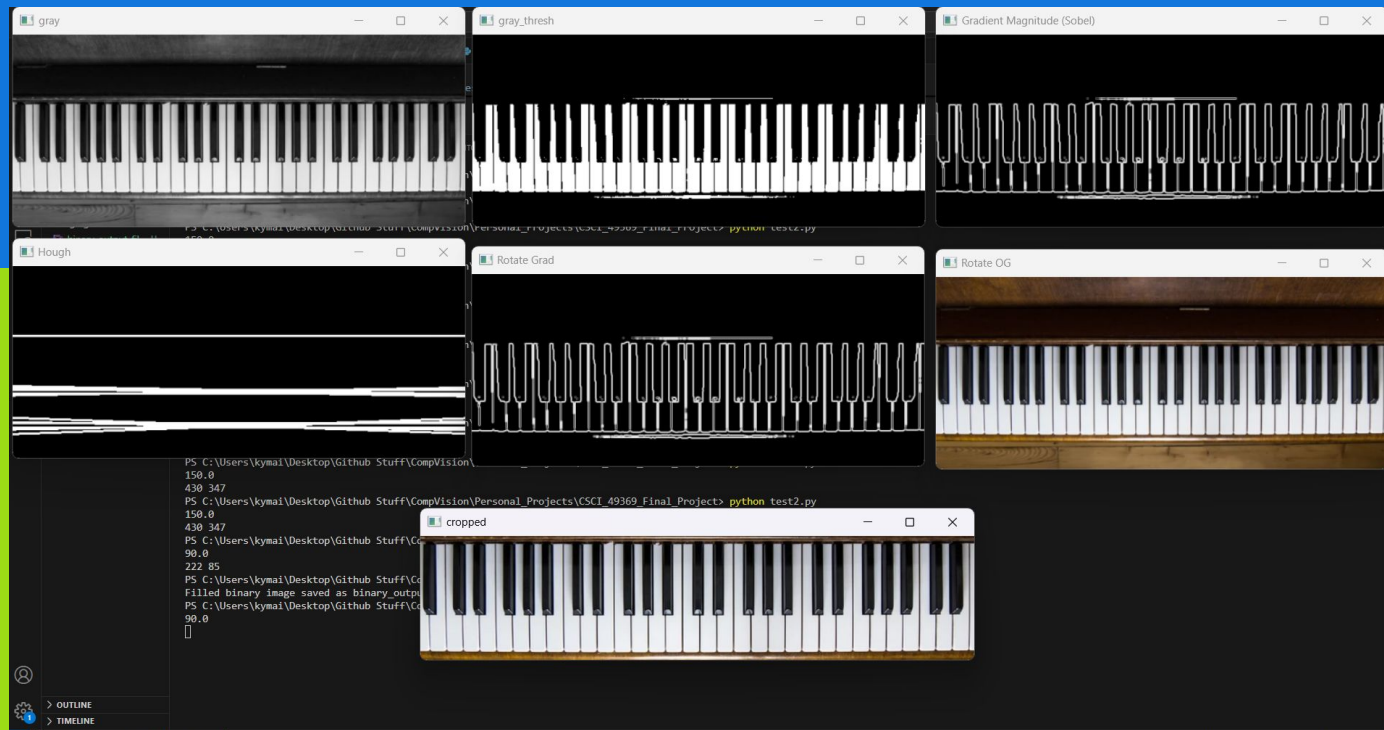
How we initially thought the problem would pan out

1. Hand detection
2. Identify black and white keys using edge detection
3. Find the locations of each key
4. See which key is being hovered / pressed

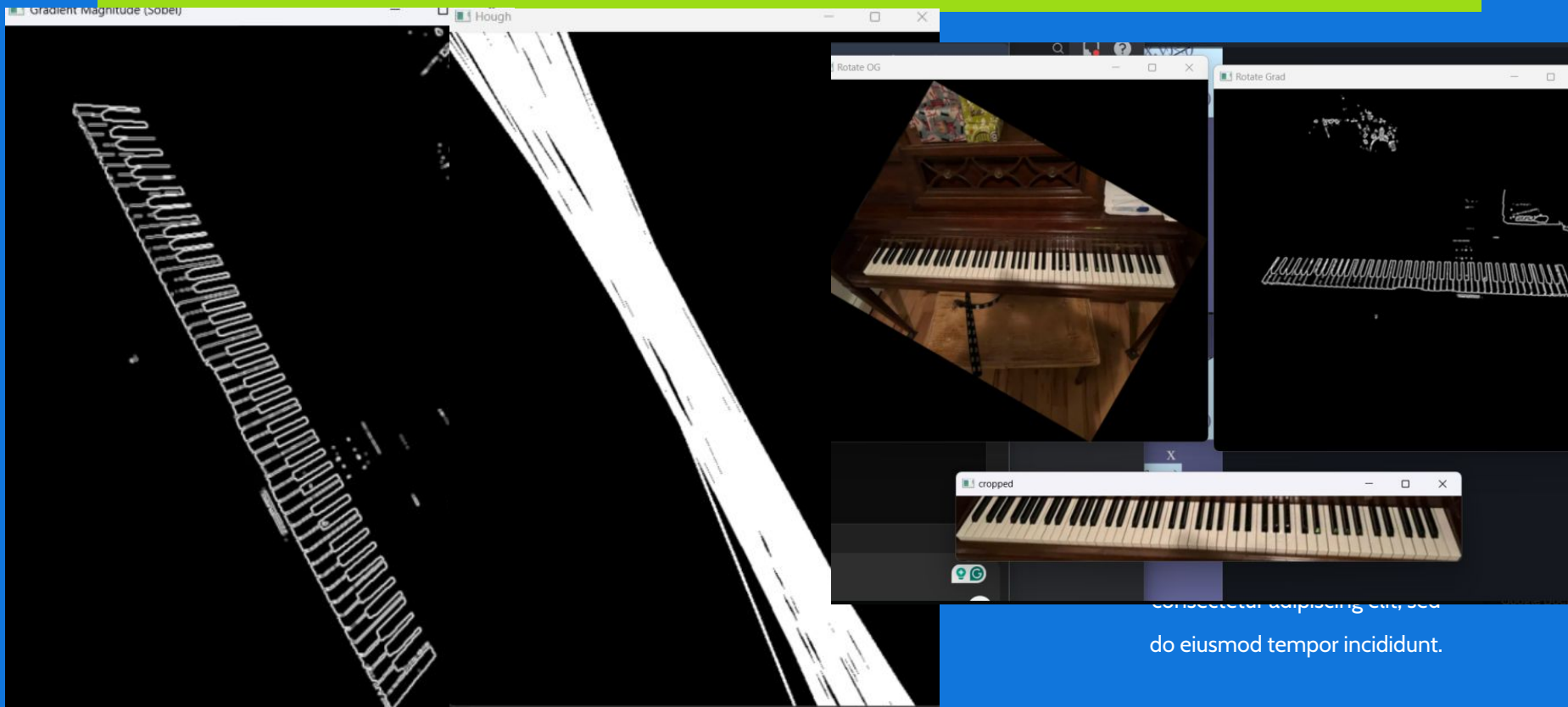
How it actually panned out

1. Hand detection
2. Cropping
3. Edge detection
4. Find only the black keys
5. Labeling them
6. Detected black keys being hovered

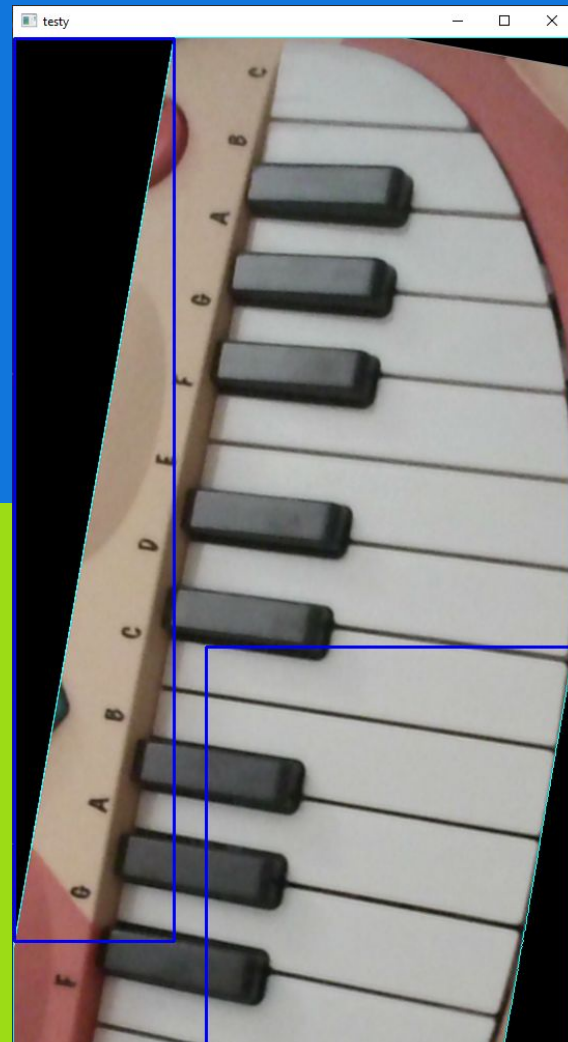
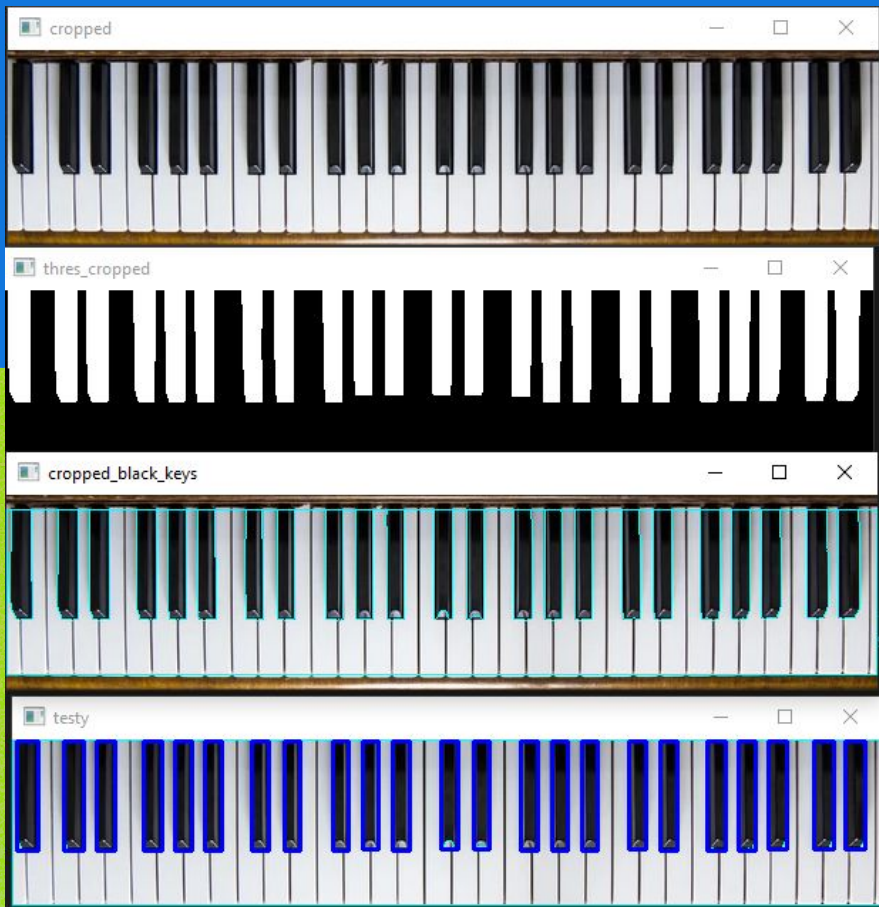
CROP THE PIANO

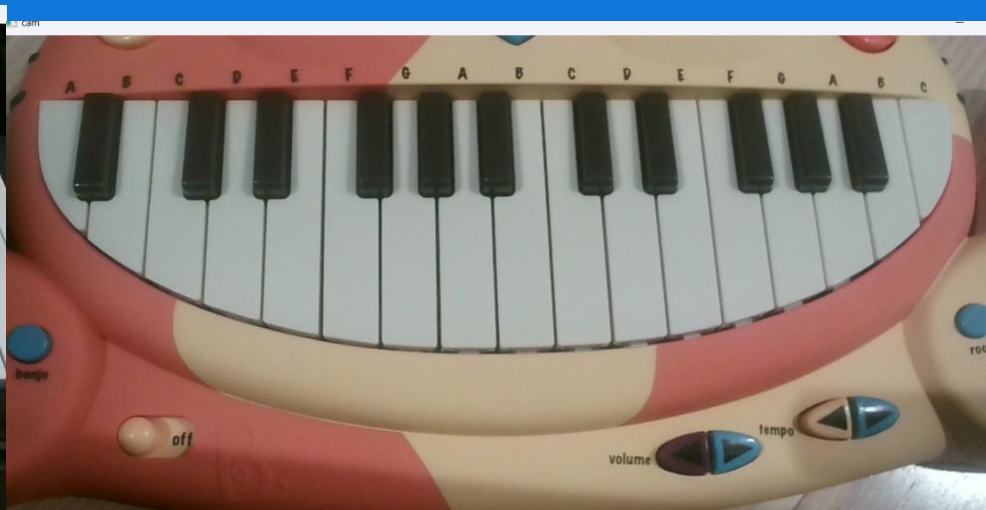
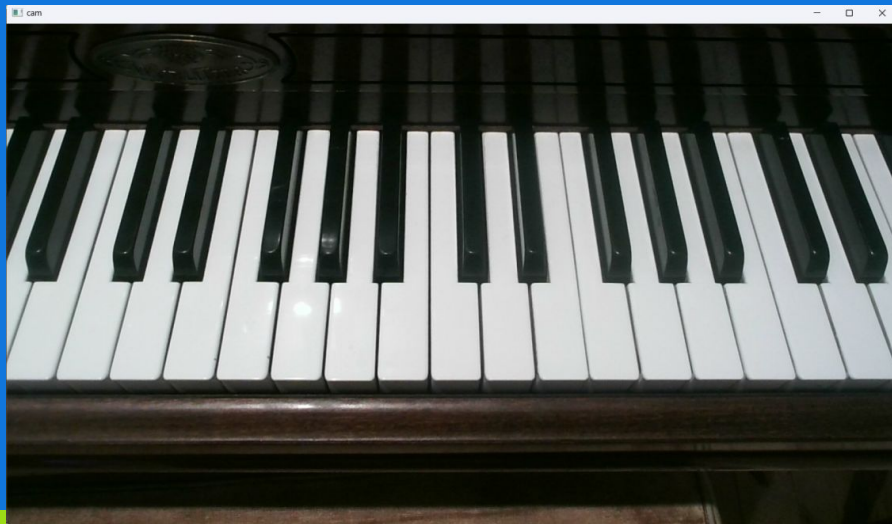


ROTATION LIMITATION



MAJOR FLAW





DEMO



FUTURE GOALS



1.

Make our program adapt to all
types of pianos



2.

Try to get as close to 100%
accuracy when detecting key
press with 2 cameras



3.

Make a music sheet with the
pressed keys

Any Questions?

